

Description of the Cleaned NHANES Dataset

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Introduction

The National Health and Nutrition Examination Survey (NHANES) is a program of the National Center for Health Statistics (NCHS), part of the U.S. Centers for Disease Control and Prevention, that assesses the health and nutritional status of adults and children in the United States (Centers for Disease Control and Prevention (CDC), National Center for Health Statistics (NCHS), n.d.). Each survey cycle combines household interviews with standardized physical examinations, including blood pressure measurements, and is designed to represent the U.S. civilian non-institutionalized population.

This report describes the cleaned NHANES dataset used in earlier assignments. It contains 29,400 participants across 3 survey waves. The aim is to summarize who is in the sample and how average systolic blood pressure is distributed, wave by wave. All figures are unweighted counts and percentages of the sample; NHANES survey weights were not applied, so the results describe this dataset rather than the U.S. population.

Data and Variables

Three variables are summarized:

- **Gender** (RIAGENDR), recorded as Male or Female.
- **Ethnicity** (ethnicity_2, derived from RIDRETH3), with six categories: Mexican, Other Hispanic, White, Black, Asian, and Other. This version was kept instead of ethnicity_1 because it reports Asian participants as a separate group.
- **Average systolic blood pressure** (mm Hg), calculated for each participant as the mean of up to four systolic readings. Participants with no readings have a missing value (7,796 participants).

The original NHANES labels are kept for consistency with the course materials. In current practice, “gender” here would be labelled *sex*, since it records biological sex, and “ethnicity” would be labelled *race*, since it describes race rather than ethnicity.

Demographic Assessment

Table 1 shows the composition of the sample in each wave. The largest wave is 2015, with 10,175 participants, and the smallest is 2017, with 9,254 participants.

The share of males ranges from 49.1% to 49.2% across waves, so the balance between males and females is very similar from one wave to the next.

Table 1: Sample characteristics of the cleaned NHANES dataset by wave

Characteristic	NHANES wave		
	2015	2016	2017
Sample size	10,175	9,971	9,254
Males, n (%)	5,003 (49.2%)	4,892 (49.1%)	4,557 (49.2%)
Ethnicity, n (%)			
Mexican	1,730 (17.0%)	1,921 (19.3%)	1,367 (14.8%)
Other Hispanic	960 (9.4%)	1,308 (13.1%)	820 (8.9%)
White	3,674 (36.1%)	3,066 (30.7%)	3,150 (34.0%)
Black	2,267 (22.3%)	2,129 (21.4%)	2,115 (22.9%)
Asian	1,074 (10.6%)	1,042 (10.5%)	1,168 (12.6%)
Other	470 (4.6%)	505 (5.1%)	634 (6.9%)

Note:

Percentages are within each wave. 'Gender' (RIAGENDR) records biological sex and 'ethnicity' (RIDRETH3) describes race; original NHANES labels are kept.

Across all waves combined, the largest ethnic group is White (33.6% of participants), followed by Black (22.1%) and Mexican (17.1%). The smallest group is Other (5.5%). Because NHANES deliberately oversamples some groups, these proportions reflect the survey design and should not be read as the ethnic composition of the U.S. population.

Blood Pressure Analysis

Table 2: Average systolic blood pressure (mm Hg) by NHANES wave

Wave	N	Average systolic BP (mm Hg)				SD
		Min	Max	Median	Mean	
2015	7528	64.7	228.7	115.3	118.3	18.1
2016	7363	74.0	231.3	117.3	120.4	18.4
2017	6713	72.7	234.0	118.0	121.7	20.3

Note:

Average of up to four readings per participant; N counts participants with at least one reading.

Table 2 summarizes average systolic blood pressure for the 21,604 participants with at least one reading. The mean is highest in wave 2017 (121.7 mm Hg) and lowest in wave 2015 (118.3 mm Hg), a difference of 3.3 mm Hg.

In every wave the mean is higher than the median, which indicates a right-skewed distribution: a minority of participants with high blood pressure pulls the mean upward. The overall range runs from 64.7 to 234.0 mm Hg. The low values largely reflect children and adolescents in the sample, whose blood pressure is naturally lower than that of adults.

Summary

The cleaned NHANES dataset includes 29,400 participants across 3 waves. The balance of males and females is broadly similar across waves, and White participants form the largest ethnic group. Average systolic blood pressure varies only modestly between waves, with means between 118.3 and 121.7 mm Hg. These descriptive results provide context for later analyses of blood pressure; such analyses should account for the wide age range in the sample and, for population estimates, the NHANES survey weights.

Reference

Centers for Disease Control and Prevention (CDC), National Center for Health Statistics (NCHS). n.d. “National Health and Nutrition Examination Survey Data.” U.S. Department of Health and Human Services. Accessed 2026. <https://www.cdc.gov/nchs/nhanes/>.